GW BME CAPSTONE · TCD PHANTOM

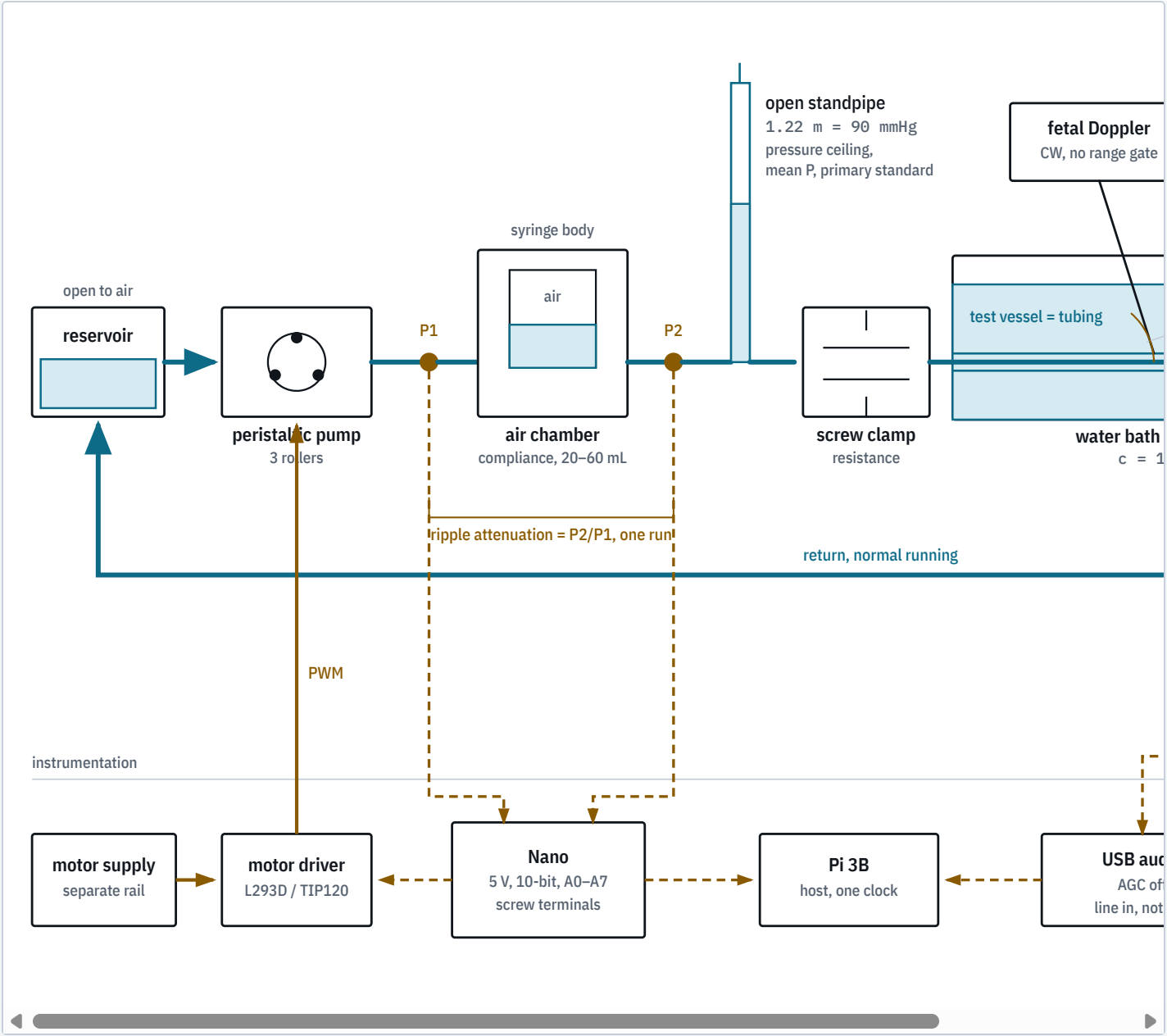
TCD Phantom Loop Layout

Two drawings. The $100 bench rig that gets built first, and the real loop it turns into once Group 1 casts a wall-less vessel. The second is not a bigger copy of the first, and the differences are where the design decisions sit.

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1 Phase 0 bench rig

Built from parts on hand. Decides the architecture before anything expensive is bought.



Two pressure taps bracketing the air chamber is the measurement. P2 over P1 gives the plenum's attenuation of the roller ripple as a ratio in a single run. With one sensor you move it and re-run, which folds pump drift into the comparison.

fluid signal power

Drawing scrolls sideways.

The pump is the device under test, not just the drive.

Ripple frequency is rollers × RPM / 60 and it tracks pump speed, never commanded heart rate. E3 is the experiment that decides whether a peristaltic belongs in the final build.

Motor on its own rail, grounds tied at one point.

The pressure sensor's output is a fixed fraction of its supply, so slow supply drift cancels when the ADC shares that 5 V reference. Fast motor transients do not cancel. Keep them apart.

The standpipe does three jobs.

An open vertical tube cannot exceed its own height, so it is the pressure ceiling. It reads mean pressure better than a ±9 mmHg transducer. And it is a primary standard, which is what you calibrate the transducers against. It will not give you a waveform.

One host owns both streams.

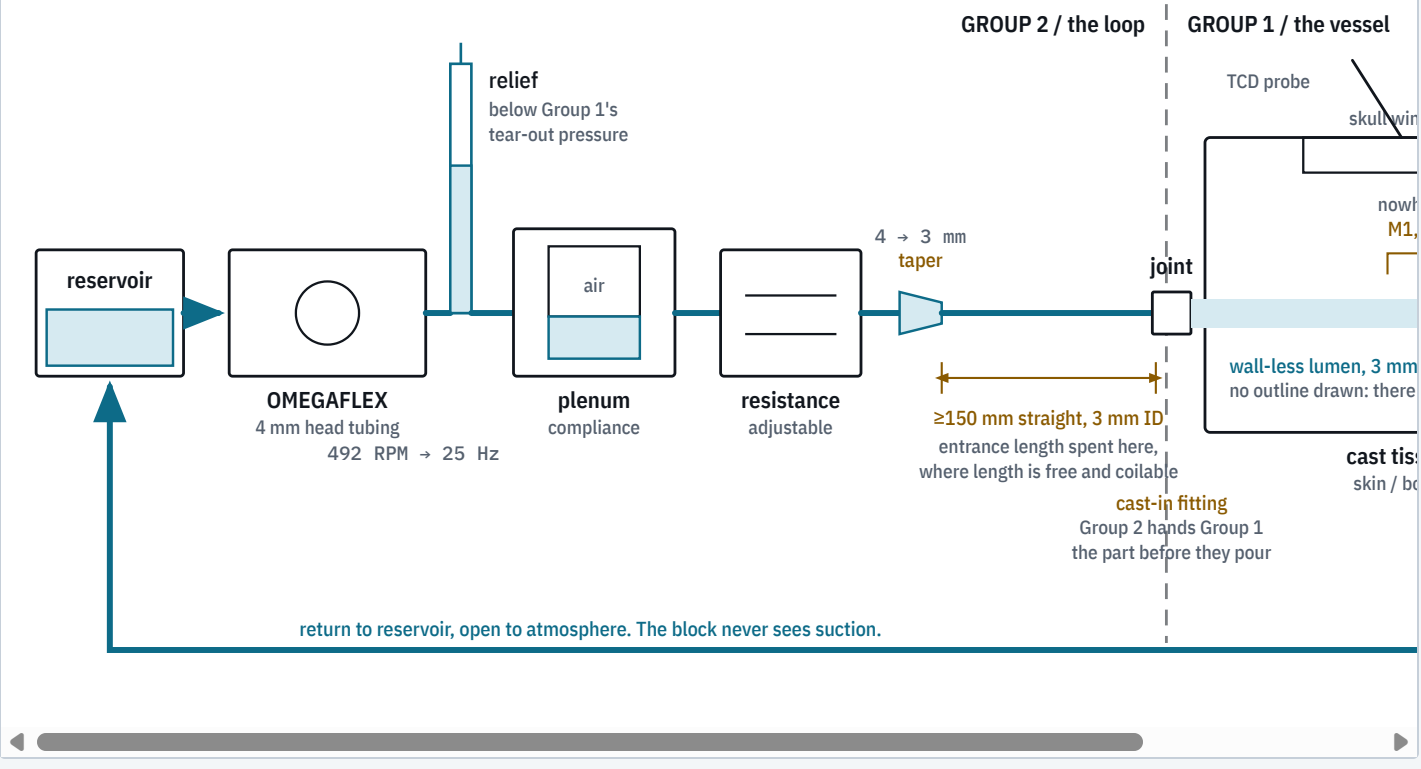
Doppler audio and pressure land on the same machine so they share a clock. Harmless for Phase 0, since PI is a ratio and the compliance sweep compares cycle averages, but it costs nothing now and saves a rebuild if anyone later wants pressure-to-velocity phase.

Commanded, actual, measured are three different numbers.

Calibration compares *measured* against *actual*, never against *commanded*. That is why the scale is in the drawing.

2 The real loop, and the seam between the teams

Same topology, different vessel. Group 1 casts a wall-less 3 mm lumen, which changes where the plumbing has to taper.



The taper happens in tubing, upstream, because the block has nowhere to put it. Flow leaving a contraction needs 80 mm to settle at 55 cm/s and 155 mm at 110 cm/s. M1 is 22 to 27 mm long inside a head-sized block. Taper at the inlet fitting and the insonated segment sits in a jet.

The fitting is the only item with a hard ordering constraint.
You cannot add a fitting to a wall-less lumen after the fact. It has to be buried in the block while it is cast, which means Group 1 cannot freeze the mold until Group 2 picks the hardware.

Pressure side only.
A wall-less lumen has no wall. Nothing resists collapse but the gel, so the block goes downstream of the pump with an open reservoir after it, and never on the suction side.

The ceiling has to be mechanical, not procedural.
The pump will deliver its full 20 psi, about 1000 mmHg, against a kink. That is roughly ten times what the phantom needs and well past what soft gel holds. One kinked line destroys days of casting.

The vessel is a compliance neither team controls.
With no wall it is the softest element in the loop and it dilates every beat, so 3 mm is a time-varying number. Any velocity computed as flow over a fixed area carries that error. Tune with the real block in the circuit, or with a compliance-matched stand-in.

3 What sets the pump

Velocity is flow over area, so the lumen diameter sets the flow requirement and nothing else does.

LUMEN DIAMETER	FLOW NEEDED FOR 110 CM/S PEAK SYSTOLIC
2.4 mm	299 mL/min
3.0 mm	467 mL/min
3.5 mm	635 mL/min

Flow follows the lumen, as the square of diameter. Group 1 owns this number, and nobody buys a pump until it is frozen.

HEAD TUBING	PUMP SPEED AT 467 ML/MIN	3-ROLLER RIPPLE	VERDICT
8.0 mm	123 RPM	6.1 Hz	sits on the systolic upstroke
6.0 mm	218 RPM	10.9 Hz	marginal
4.0 mm	492 RPM	24.6 Hz	clear of the heartbeat, filterable

Ripple follows the head tubing alone, here at a fixed 467 mL/min. These are two independent choices, which is why they are two tables: tapering after the pump changes the flow the vessel sees and leaves the ripple exactly where it was. Figures are scaled from one published point, 2280 mL/min at 600 RPM on 8 mm, assuming flow goes as the square of bore. Check them against the real curve, and count the rollers on the head.

Drawn for Group 1 and Group 2, October 2026. Boundary and jointly owned specs live in `capstone/INTERFACE-CONTROL-DOCUMENT.md`. If a number here disagrees with that file, that file wins, and tell me.